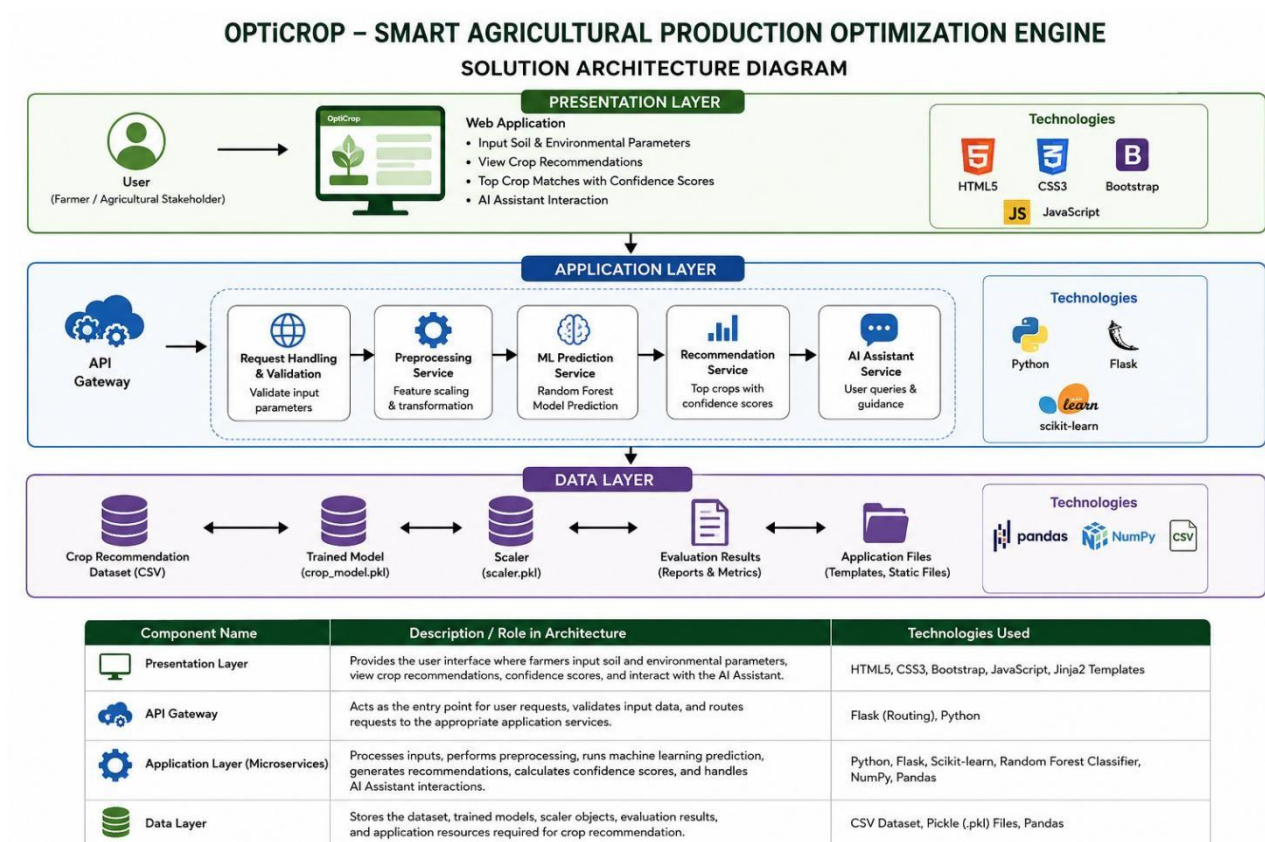


Solution Architecture

Date	25 June 2026
Team ID	XXXXXX
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface where farmers enter soil and environmental parameters, view crop recommendations, confidence scores, and interact with the AI Assistant.	HTML5, CSS3, Bootstrap 5, JavaScript, Jinja2 Templates
API Gateway	Acts as the entry point for user requests, receives input data from the frontend, validates parameters, and routes requests to the machine learning prediction services.	Flask Routing, Python
Microservices	Processes user inputs, performs feature scaling, executes machine learning predictions, calculates confidence scores, generates top crop recommendations, and manages AI Assistant interactions.	Python 3.11, Flask, Scikit-learn, Random Forest Classifier, NumPy, Pandas
Database	Stores the agricultural dataset, trained machine learning models, preprocessing artifacts, evaluation results, and application resources required for crop recommendation.	CSV Dataset, Pickle (.pkl) Files, Crop Recommendation Dataset, Scaler Model

- **Presentation Layer:** The Presentation Layer provides the user interface of the OptiCrop system, allowing farmers and agricultural users to enter soil nutrient values and environmental parameters, view crop recommendations, confidence scores, and interact with the AI Assistant through a responsive and user-friendly web application.
- **Application Layer:** The Application Layer serves as the core processing unit of the system. It handles user requests, validates input data, performs feature scaling, executes machine learning predictions using the Random Forest model, generates confidence scores and alternative crop recommendations, and manages communication between the frontend and data layer.
- **Data Layer:** The Data Layer stores and manages all data resources required by the system, including the Smart Agriculture Crop Recommendation Dataset, trained machine

learning models, preprocessing artifacts, evaluation results, and supporting files used during prediction and analysis

- **Overall Architecture:** The layered architecture of OptiCrop ensures efficient data flow, modularity, scalability, maintainability, and seamless integration of machine learning capabilities. It enables accurate crop recommendations while providing a strong foundation for future enhancements such as weather forecasting, fertilizer recommendation, yield prediction, IoT integration, and mobile application support.